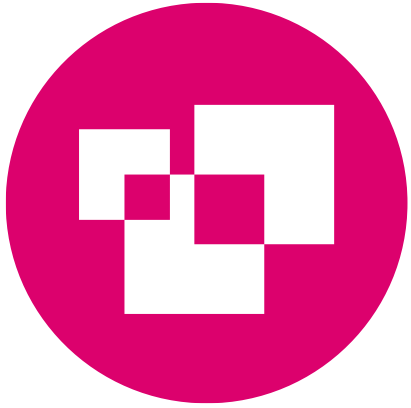




World
Weather
Attribution

Climate Justice outside UNFCCC

Prof Fredi Otto

Centre for Environmental Policy
Imperial College London

[@frediotto.bsky.social](https://bsky.app/profile/frediotto.bsky.social)



What is Climate Justice?

When we talk about climate change...

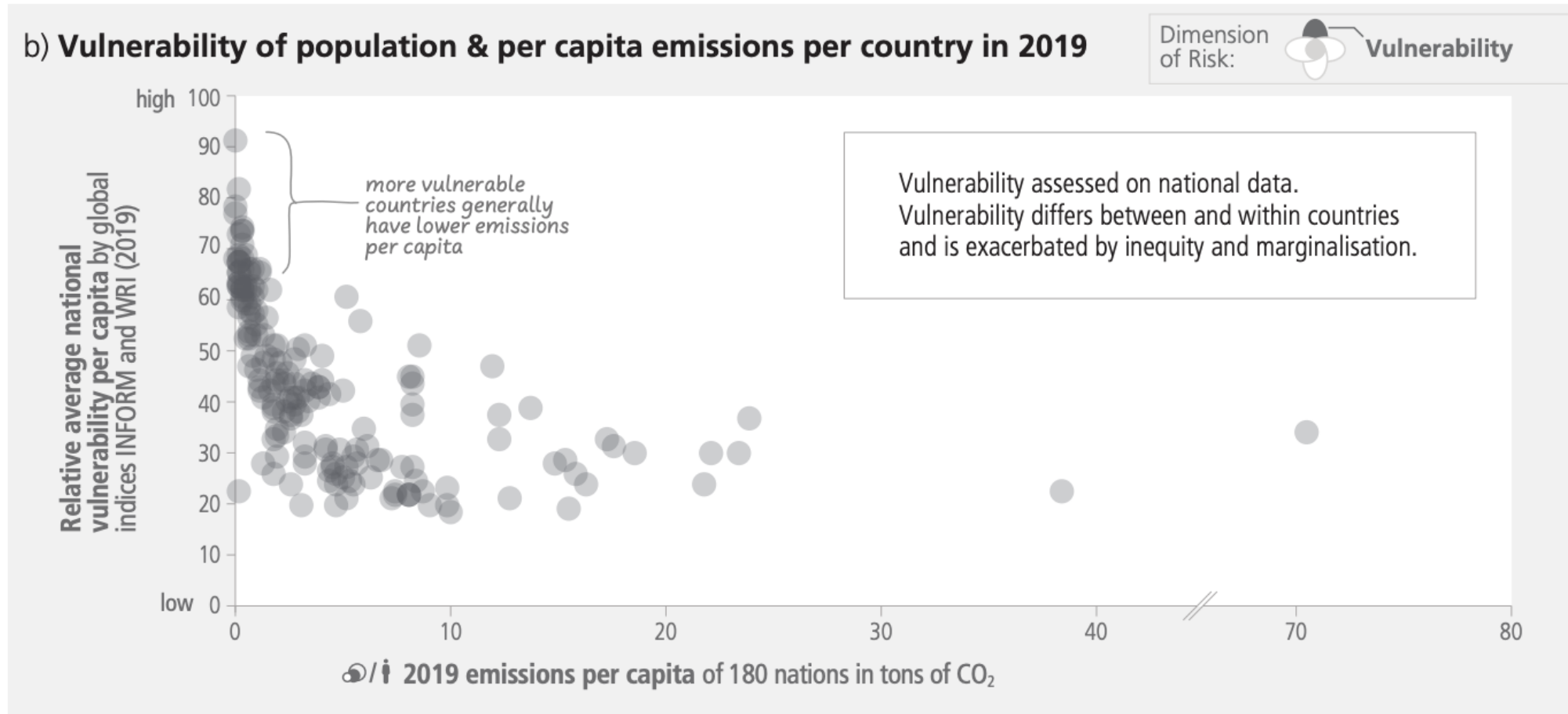


<https://www.worldweatherattribution.org> @ClimateAdam

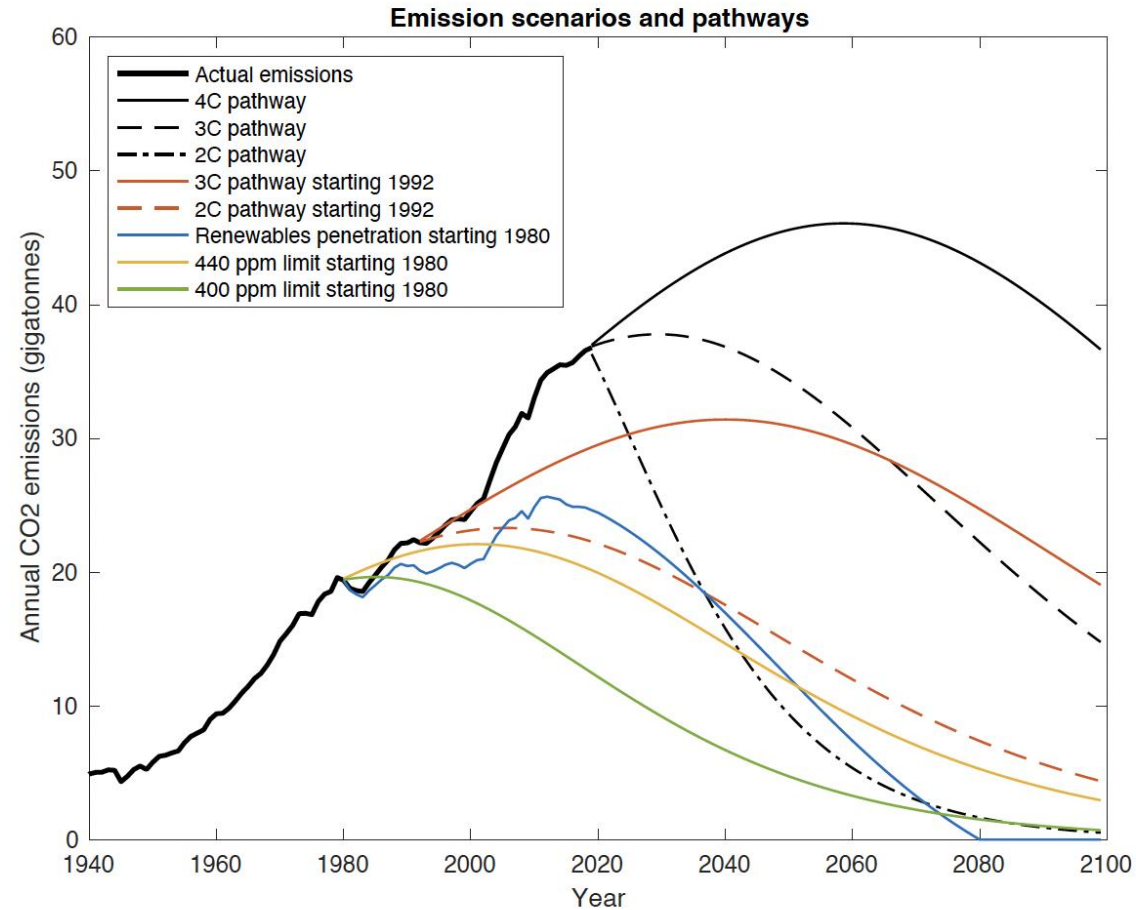


Losses and damages – now and in the future

Distributive justice

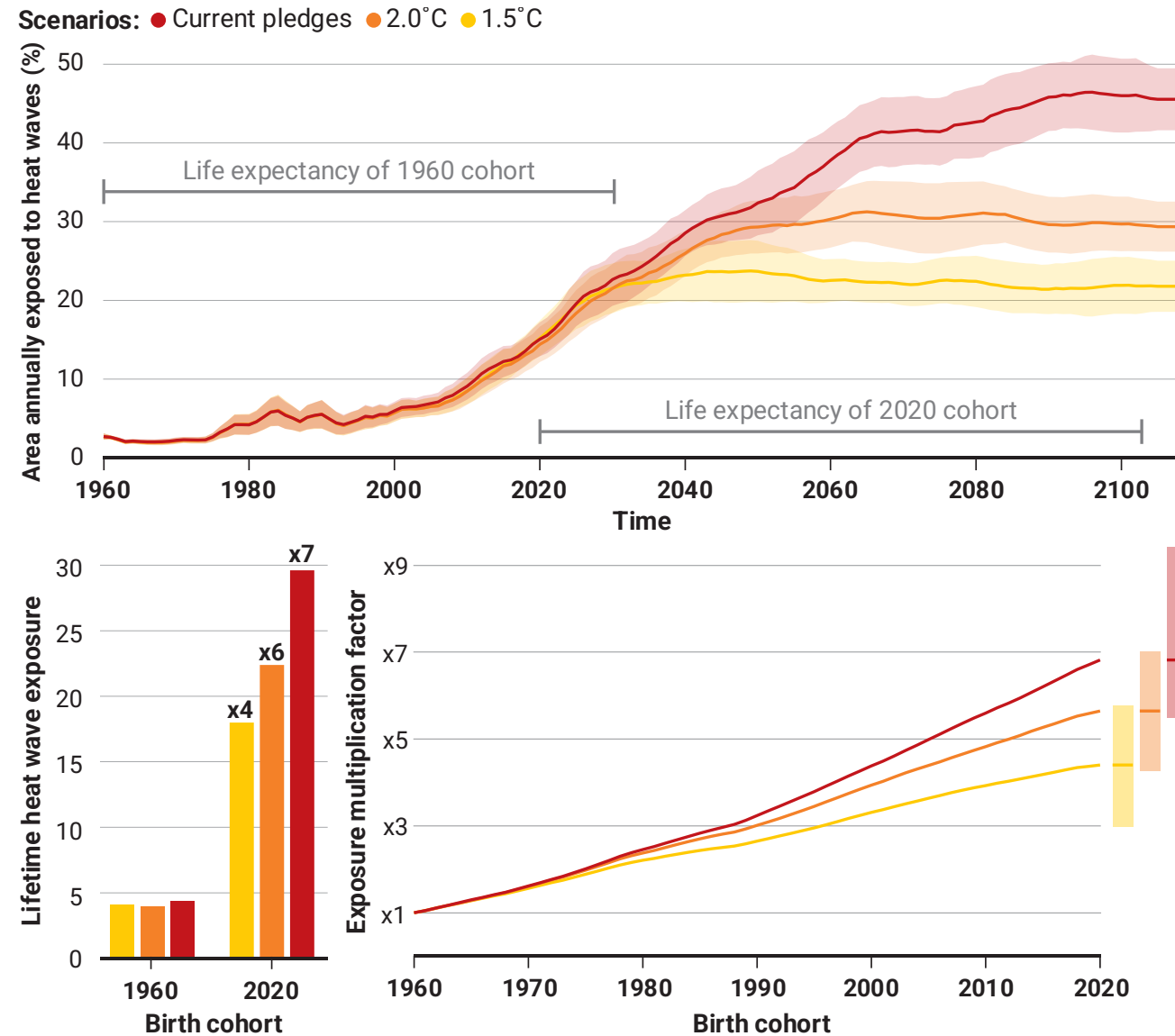


Intergenerational justice



“Uncertainties in the costs are high, but the annual increase in cost is relatively well constrained - with each additional year of delay in implementing mitigation costing an additional 0.3–0.9 trillion dollars in total discounted future mitigation costs, if the 2C target is to be ultimately met.” *Sanderson & O’Neill 2020*, <https://doi.org/10.1038/s41598-020-66275-4>

Intergenerational justice



(Historic) responsibility



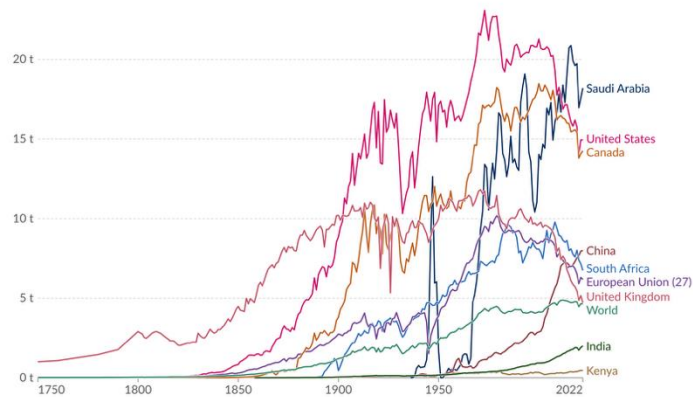
Climate change is fundamentally a political issues creating “winners” and “losers”

Principles to consider:

- Ability to pay
- Beneficiary pays
- Polluter Pays
- Do No harm

Per capita CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry¹. Land-use change is not included.



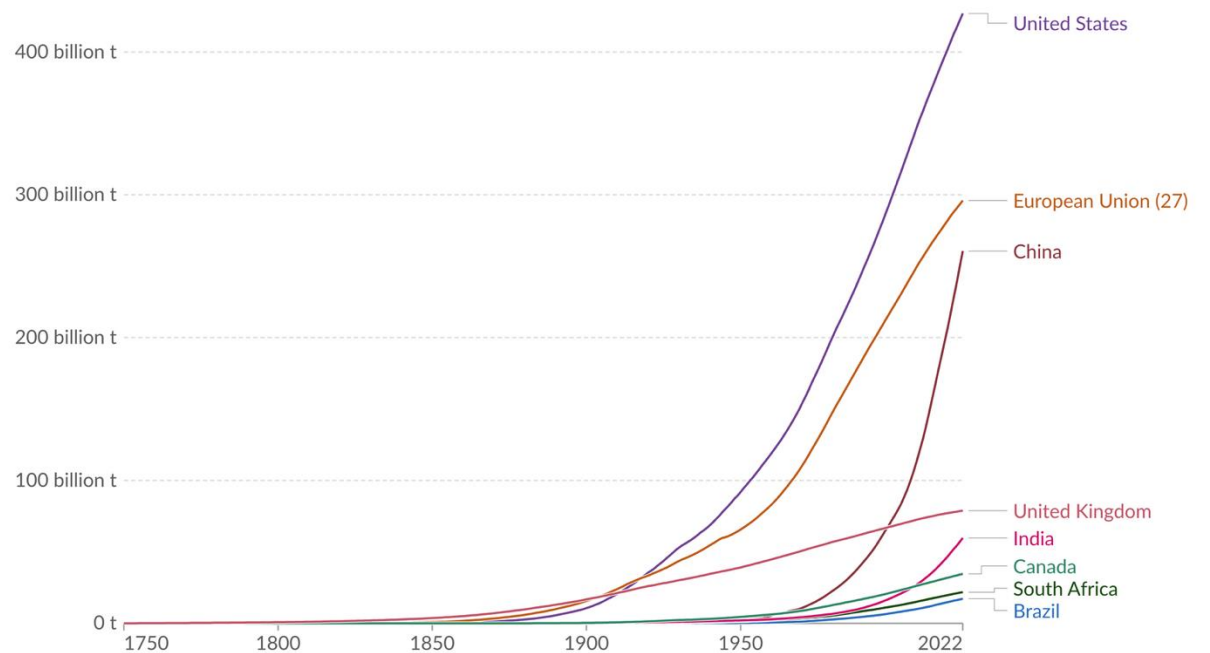
Data source: Global Carbon Budget (2023); Population based on various sources (2023)
OurWorldInData.org/co2-and-greenhouse-gas-emissions | CC BY

1. Fossil emissions: Fossil emissions measure the quantity of carbon dioxide (CO₂) emitted from the burning of fossil fuels, and directly from industrial processes such as cement and steel production. Fossil CO₂ includes emissions from coal, oil, gas, flaring, cement, steel, and other industrial processes. Fossil emissions do not include land use change, deforestation, soils, or vegetation.

Cumulative CO₂ emissions

Running sum of CO₂ emissions produced from fossil fuels and industry¹ since the first year of recording, measured in tonnes. Land-use change is not included.

Our World
in Data



Data source: Global Carbon Budget (2023)

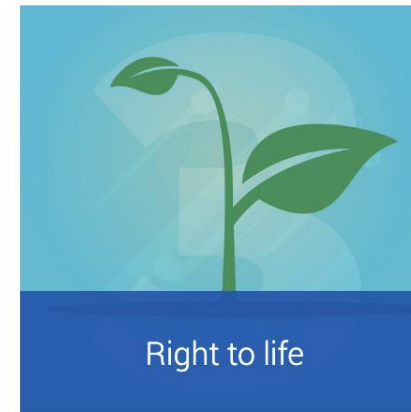
OurWorldInData.org/co2-and-greenhouse-gas-emissions | CC BY

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Why does all this matter? – climate justice



Acknowledging that climate change is a common concern of humankind, Parties should, when taking action to address climate change, respect, promote and consider their respective obligations on human rights, the right to health, the rights of indigenous peoples, local communities, migrants, children, persons with disabilities and people in vulnerable situations and the right to development, as well as gender equality, empowerment of women and intergenerational equity. (Preamble, Paris Agreement)



“Despite strong references to justice, human rights, and equity in the Paris Agreement’s preamble, the concept of equity is largely absent from its substantive components.”

Klinsky et al., 2017 <https://doi.org/10.1016/j.gloenvcha.2016.08.002>



Impacts on equality, equity...

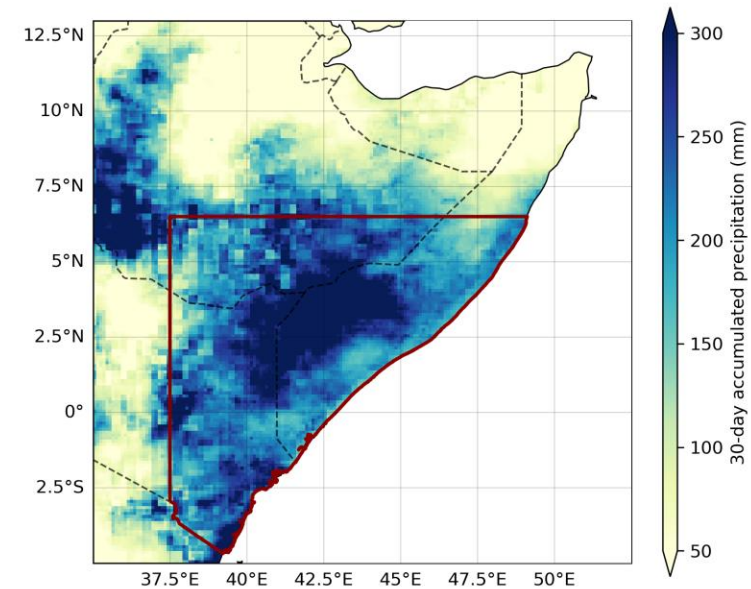
Every new extreme hits the poorest harder & harder – increasing already high inequality

- Kenya & East Africa has been facing disaster after disaster, including prolonged drought between 2020-23, and multiple episodes of torrential rainfall leading to severe flooding.
- Kenya has the highest number of refugees globally
- Rapid urbanisation is amplifying flood risks, especially in large informal areas that are located on flood-prone land
- lack adequate structural protections from the rains, and whose residents lack resources to recover and rebuild.
- Land-use changes, including deforestation and conversion to agricultural land are also occurring to different degrees in each of the countries studied, adding to flood risk.

Kimutai et al., 2025

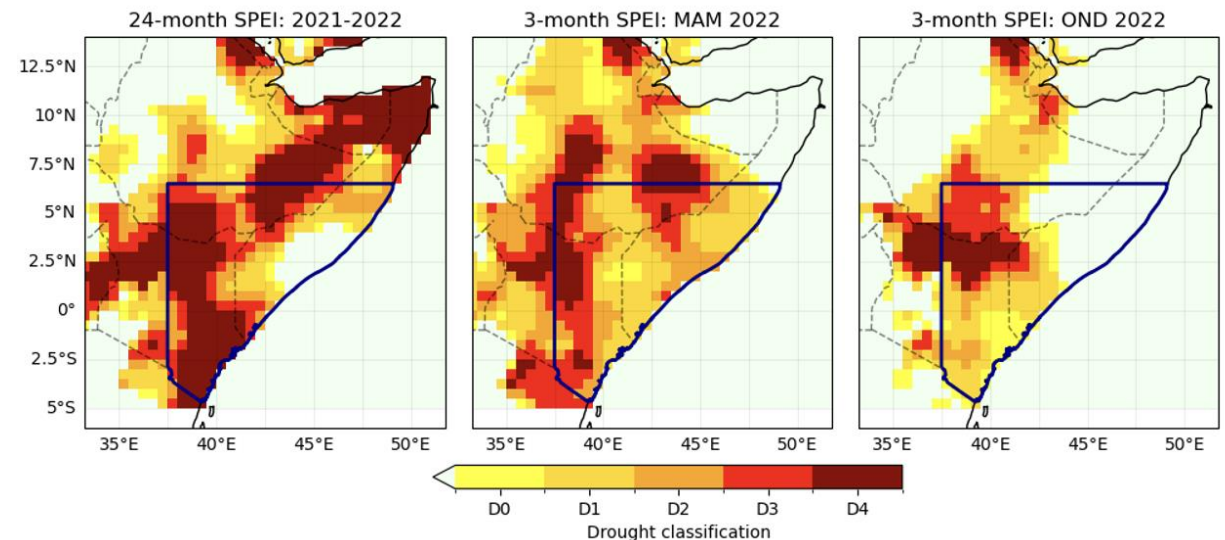
<https://www.sciencedirect.com/science/article/pii/S2212094725000039>

Flood 2023

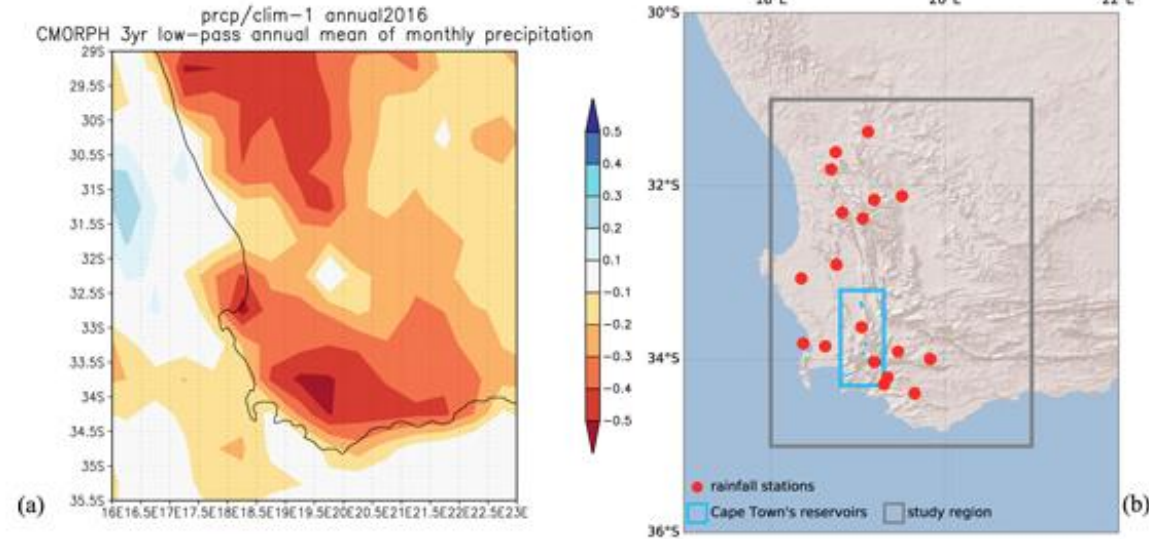


Drought 2021/22

CPC drought classifications using standardised precipitation-evapotranspiration index (SPEI)



Cape Town Drought - Racism

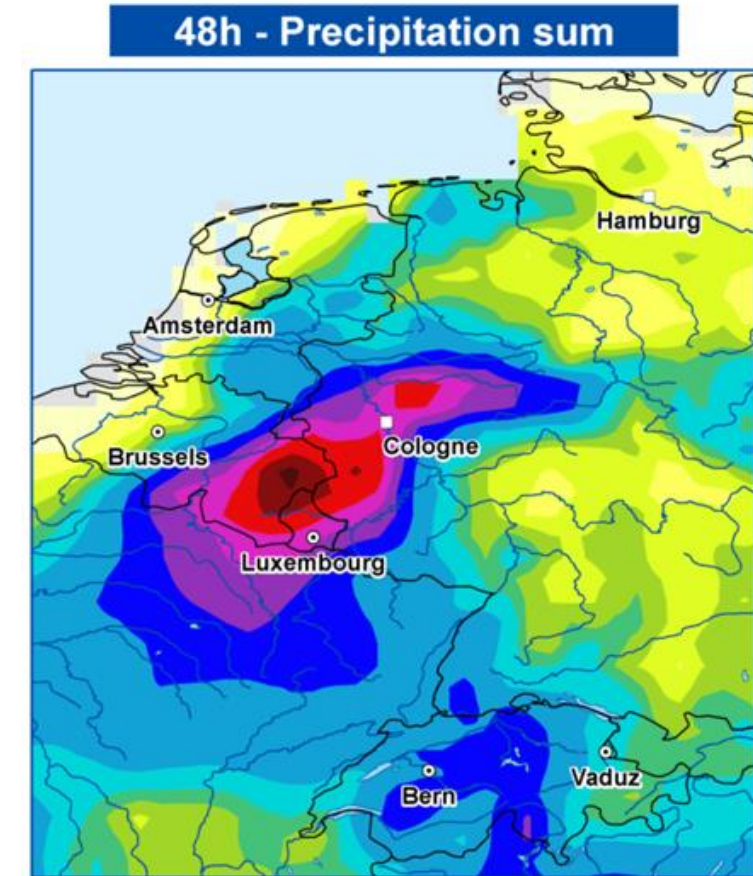


- Avoided day zero through strict water restrictions
- No irrigation in fields and gardens/plots – agricultural workers losing their jobs, families using garden produce for subsistence lose their income
- Water restrictions enforced in predominantly black areas
- Predominantly white areas use groundwater wells, swimming pools and tanks to avoid restrictions
- Originally very good water management, populist policy change led to stalling of plans

Germany – lack of early warning/ education/ arrogance



- Over 200 deaths despite early warning - warnings did not reach the population in affected area
- 1/3 did not know anything was going to happen
- Less than 10% was expecting danger to life
- 50% of those that heard warnings didn't know the cause of action
- no warnings tailored to vulnerable groups – many people who died were those with mobility or educational difficulties
- Post disaster assessments hindered by stations that had been destroyed in the floods and reluctance of agencies to address questions of vulnerability and exposure



Precipitation data: Extended version of E-OBS. Graphic credits: © Deutscher Wetterdienst
Geodata: © GeoBasis-DE/BKG 2020 (Last update: 01.01.2020).

Pakistan – Colonialism, lack of learning



- 1730 lives lost, 33mio people affected, Pushed between 8.4 and 9.1 million more people below the poverty line.
- Over 30bn \$ losses more than 16bn \$ needed for immediate reconstruction
- Outdated colonial river management systems (that were never really fit for purpose)
- Building in floodplains
- Lessons learned from previous large scale flooding in 2010 not implemented

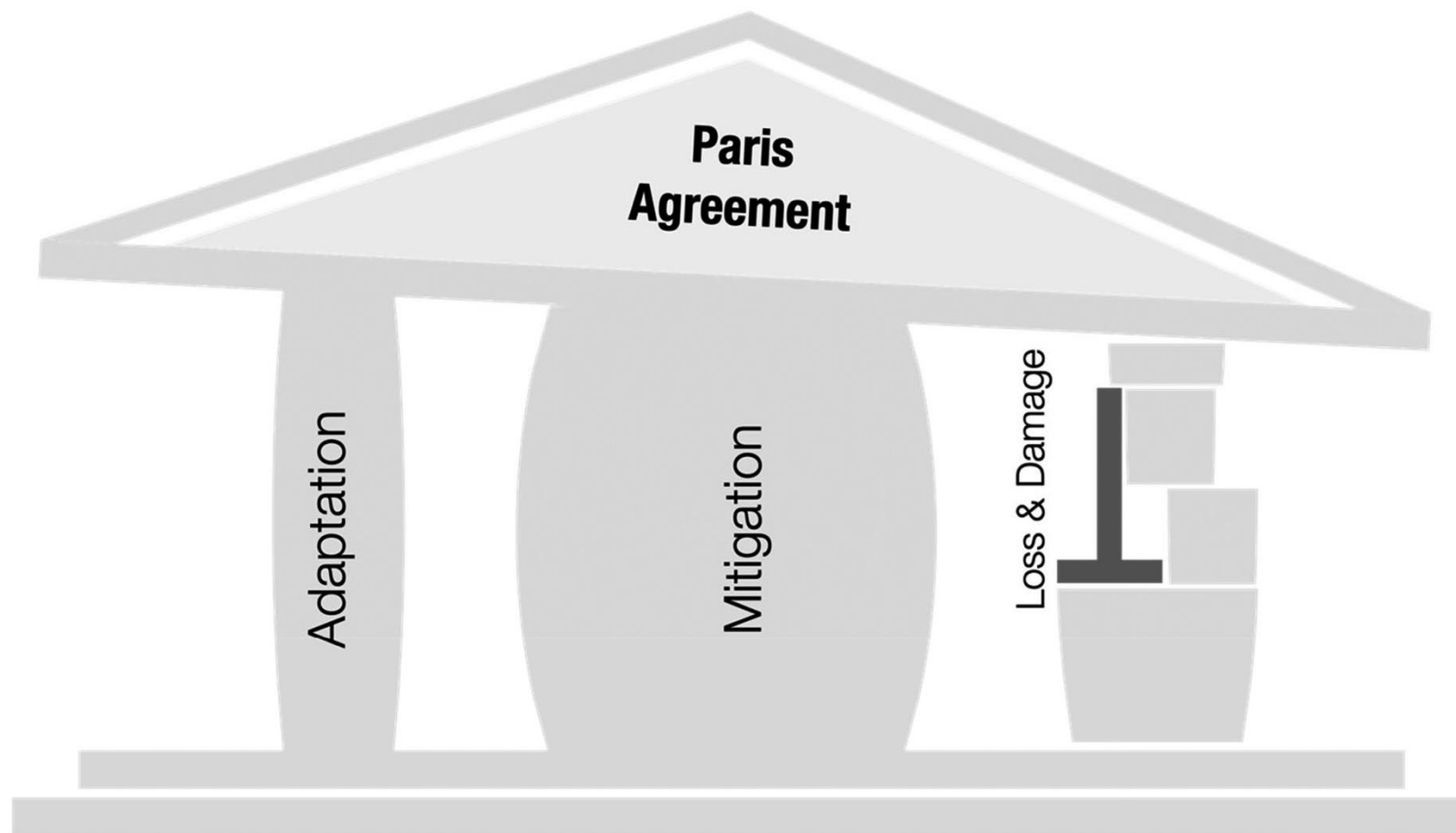


Loss & Damage

one response

The Paris Agreement

the reality (optimistic version)





Addressing L&D outside the UNFCCC



- People are increasingly turning to courts for climate remedies due to the mounting severity of the climate crisis and the **striking inadequacy** of public and private sector responses.
- cases are **strategic lawsuits aimed at achieving broad social impacts** beyond protecting the interests of individual claimants
- The opportunities and prospects for such lawsuits have improved as researchers have generated **new evidence** on the harmful effects of climate change, the emissions contributions of government and corporate entities, and the scope of efforts undertaken by **bad faith actors** to sow doubt about climate science and delay climate action.
- The importance of litigation goes beyond final court decisions. As with tobacco litigation, climate litigation can provide a pathway for uncovering information about corporate responsibility for climate change, affecting corporate reputations in the court of public opinion and hence, their **“social license”** to operate in ways that harm public health and the environment.

Industry disinformation on global warming



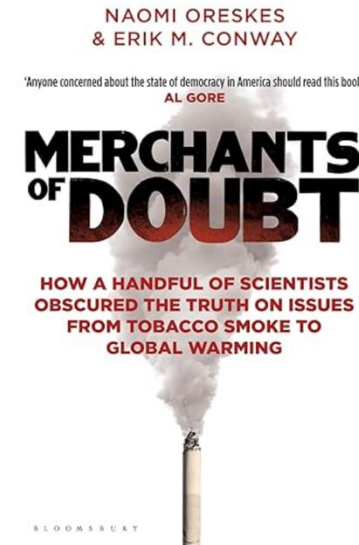
Addressed to the
American Petroleum
Industry Association (API)
in 1954

Perhaps the most interesting effect concerning carbon in trees which we have thus far observed is a marked and fairly steady increase in the C^{12}/C^{13} ratio with time. Since 1840 the ratio has clearly increased markedly. This effect can be explained on the basis of a changing carbon dioxide concentration in the atmosphere resulting from industrialization and the consequent burning of large quantities of coal and petroleum. If this explanation were correct, the carbon dioxide content of the atmosphere today would be about 5% greater than it was a century ago.

API president in 1965.

This report unquestionably will fan emotions, raise fears, and bring demands for action. The substance of the report is that there is still time to save the world's peoples from the catastrophic consequence of pollution, but time is running out.

One of the most important predictions of the report is that carbon dioxide is being added to the earth's atmosphere by the burning of coal, oil, and natural gas at such a rate that by the year 2000 the heat balance will be so modified as possibly to cause marked changes in climate beyond local or even national efforts. The report further states, and I quote: "... the pollution from internal combustion engines is so serious, and is growing so fast, that an alternative nonpolluting means of powering automobiles, buses, and trucks is likely to become a national necessity."



Franta 2018, <https://www.nature.com/articles/s41558-018-0349-9>



Addressing Loss and Damage in Court

- A state can bring a claim against another for state responsibility
- Under general international law: 'No harm rule' + procedural obligations (Environmental Impact Assessment + due diligence)
- Obligations under FCCC/Kyoto Protocol/Paris Agreement (but fuzzy obligations? Obligations of conduct etc)
- Private parties can bring claims against their own state or another state for climate-related loss and damage in international forums, if internationally protected human rights are implicated
- Private parties can bring claims against their own state or other private parties in domestic judicial forums





-
- Supreme Court of the Netherlands held that the national government had a legal obligation to protect the human rights of its citizens against the harmful impacts of climate change, and as a result, the government must take steps to reduce carbon emissions consistent with limiting global warming to 1.5°C.
 - The Court found that GHG emissions under the direct influence of Dutch national policy (approximately 0.5% global emissions) were sufficiently substantial for adjudication, that climate change endangered the health and well-being of Dutch citizens in the aggregate, and that a causal nexus existed between national policy and those injuries based on information in IPCC assessments. It was not necessary for plaintiffs to prove that the Netherlands' national contribution to climate change had caused particularized harm to specific individuals or entities because Dutch law allows organizations to file lawsuits on behalf of collective interests.

Recent successes 1

and open research questions for science/law

- **Finch v Surrey County Council** – UK supreme court clarified all emissions need to be included in EIA, based on causal arguments: “Expressed in terms of **necessary and sufficient conditions**, the “but for” test is satisfied: but for the extraction of the oil, the oil would stay in the ground and so would not be burnt as fuel. On the agreed facts, the extraction of the oil is not just a necessary condition of burning it as fuel; it is also sufficient to bring about that result because it is agreed that extracting the oil from the ground guarantees that it will be refined and burnt as fuel. As discussed above, **a situation where X is both necessary and sufficient to bring about Y is the strongest possible form of causal connection - much stronger than is required as a test of causation for most legal purposes.**”

➡ No more new extraction!?

Key oil project must count full climate impact - court



Controversial UK oil field publishes full scale of climate impact



<https://www.bbc.co.uk/news/articles/ce3xzgdqw3ro>

Recent successes 2

and open research questions for science/law

Luciano Lliuya v RWE – death by detail, BUT liability established

- Just because you have a permit for your economic activity doesn't mean you can cause harm
- small(ish) emissions are no excuse to shirk responsibility
- the climate system is not too complex to establish causality
- you are responsible to educate yourself about potential harms of your business

➡ litigation risk becoming serious?

<https://rwe.climatecase.org/en>

I-5 U 15/17

2 O 285/15
Landgericht Essen



Verkündet am 28.05.2025

Teiner, Justizsekretär
als Urkundsbeamter der
Geschäftsstelle

Oberlandesgericht Hamm

IM NAMEN DES VOLKES

Urteil

In dem Rechtsstreit

des Herrn Saúl Ananías Luciano Lliuya, [REDACTED]
[REDACTED] Huaraz,
Peru,

Klägers und Berufungsklägers,

g e g e n

RWE AG, vertreten durch den Vorstand [REDACTED]
[REDACTED]

Beklagte und Berufungsbeklagte,

hat der 5. Zivilsenat des Oberlandesgerichts Hamm
auf die mündliche Verhandlung vom 19.03.2025
durch den Vorsitzenden Richter am Oberlandesgericht [REDACTED] den Richter am
Landgericht [REDACTED] und die Richterin am Oberlandesgericht [REDACTED]

für Recht erkannt:

Die Berufung des Klägers gegen das am 15.12.2016 verkündete Urteil des
Landgerichts Essen (Az.: 2 O 285/15) wird zurückgewiesen.

Recent successes 3

and open research questions for science/law



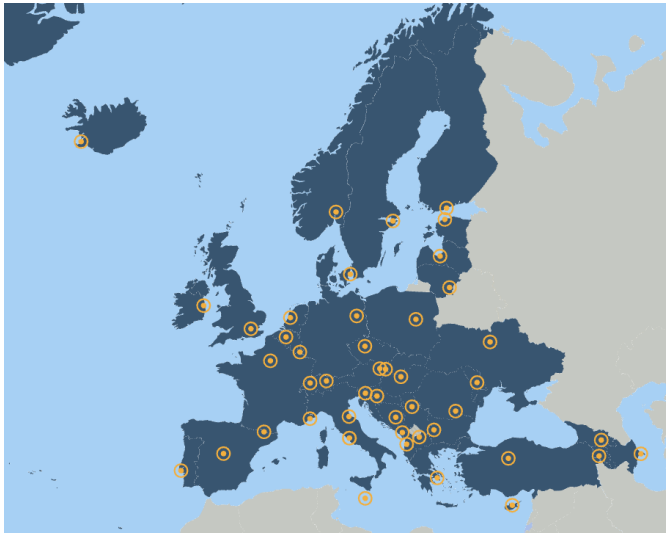
- **KlimaSeniorinnen v Switzerland** - European Court of Human Rights (ECtHR) declaring that Switzerland violated human rights (in particular the right to respect for private and family life (article 8), requiring Switzerland to calculate a fair carbon budget & a plan to stick to it.



Switzerland ignores the ruling. Can the implementation review make it stick?

Passed to Committee of Ministers (CoM) for supervision & implementation

Council of Europe
Member States



<https://www.ejiltalk.org/another-brick-in-the-wall-of-klimaseniorinnen-the-committee-of-ministers-september-2025-monitoring-decision/>

“The CoM’s stance is deeply concerning. By conflating what Switzerland *plans* to emit with what it is *permitted* to emit, the Swiss government’s approach reduces the exercise of quantification to a projection of business-as-desired. This is akin to calculating a budget by extrapolating from spending habits rather than from the actual funds available – or from the obligations to share those funds fairly with others. The result is a “budget” in name only: one that reflects Switzerland’s preferences and political feasibility, rather than the physical limits of the atmosphere or the normative limits articulated by the Court.”



437. In terms of the operation of this legal standard in the climate change context, the Court observes that causation involves two distinct elements. First, whether a given climatic event or trend can be attributed to anthropogenic climate change; and second, to what extent damage caused by climate change can be attributed to a particular State or group of States. While the second element must be established in concreto in respect of specific claims brought by States in respect of damage, in many cases the first element may be addressed by recourse to science. The scientific evidence adduced in these proceedings establishes that significant harm to the climate system and other parts of the environment has been caused as a result of anthropogenic GHG emissions (see paragraph 277 above). For example, the IPCC's 2023 Synthesis Report includes data clearly linking the human contribution to climate change to observed increases in heatwaves, flooding and drought (p. 48, figure 2.3).

409.Thus, a State that does not exercise due diligence in the performance of its primary obligation to prevent significant harm to the environment, including to the climate system, commits an internationally wrongful act entailing its responsibility

450. Article 31 of the International Law Commission Articles on State Responsibility provides that a responsible State is under an obligation to make full reparation for the damage caused by the internationally wrongful act

- finding a forum with jurisdiction
- establishing the existence of clear and binding obligations – (much easier since July!)
- establishing breach of such obligations

Advisory opinion African Court on Human and Peoples' Rights

8. (of 10) Responsibilities of African States regarding violations by third parties (international monopolies, multinational corporations) operating within their jurisdiction. These entities contribute significantly to emissions and States have obligations under the Charter (Article 21) and international law to regulate and monitor their activities. The importance of Environmental Impact Assessments (EIAs) in regulating third-party activities is highlighted; (Pending)

Plan B. Earth and Others v United Kingdom

Whether the UK has breached obligations under the European Convention on Human Rights by failing to reduce GHG emissions. (case declared inadmissible in 2023 because claimants “not sufficiently affected”).



- Establishing damage & adaptation (there are no metrics)
- Limited understanding of evidence
- “Social license” to destroy lives of many for profits of few
- Status quo is best of all possible worlds???

Also

- Case by case rather than holistic approach
- Reactive rather than preventive or supportive
- Can play a complementary role to the multilateral regime
- Enforcement??

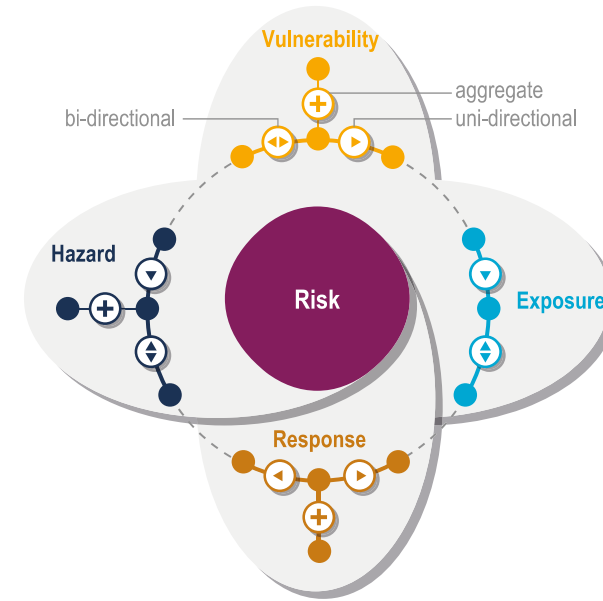
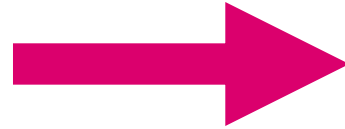




Break



Exam



From lecture 3

- What are the definitions of the components of risk?
- Why the fourth blade on responses?
- Is the fourth blade necessary?
- What are drivers of the components of risk?
- What are the ways risks can interact?
- What are examples of the different ways of how risks can interact?

From Lecture 5

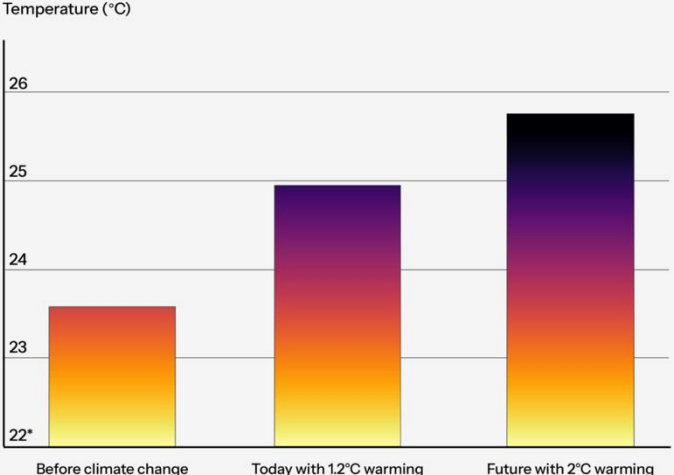
The main idea of event attribution



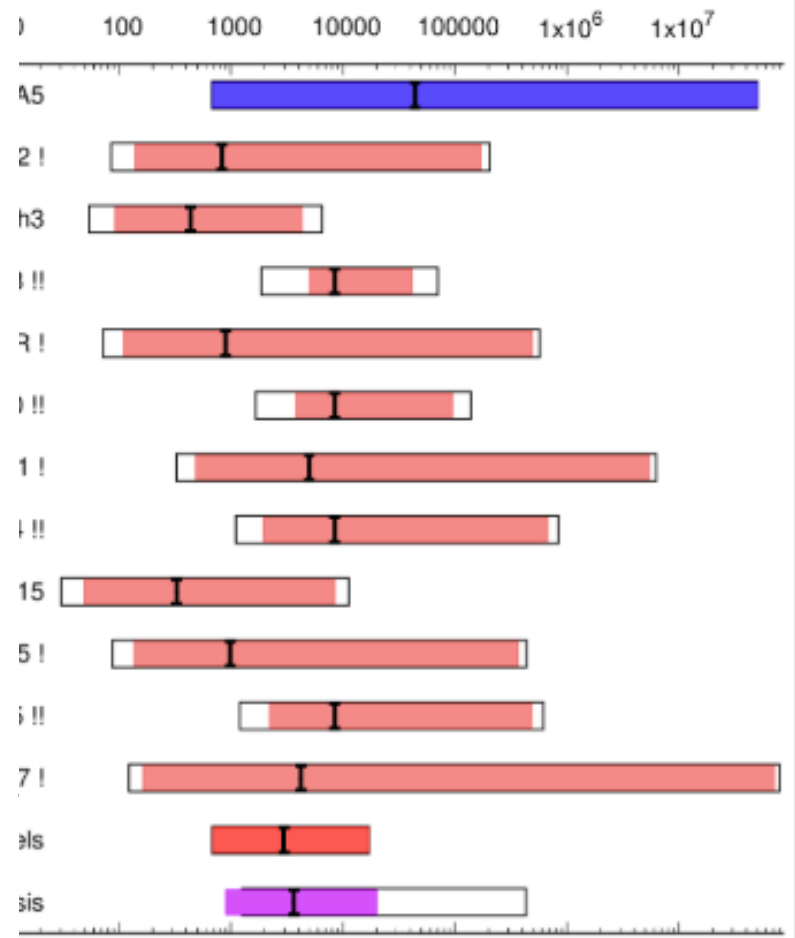
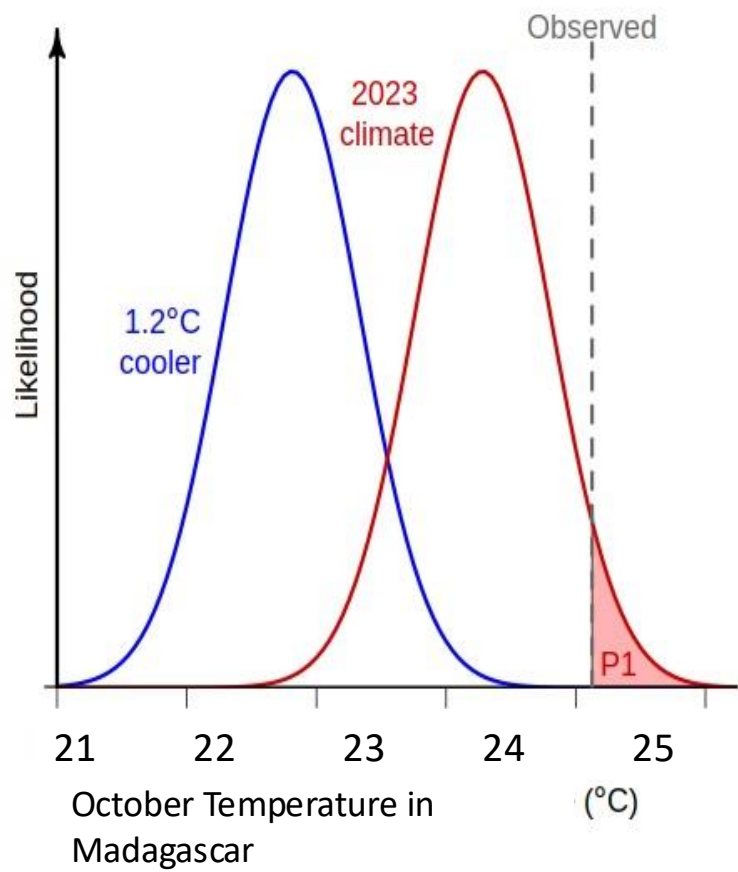
How often should we expect similar October heatwaves in Madagascar?



How has climate change influenced the average October temperature in Madagascar?

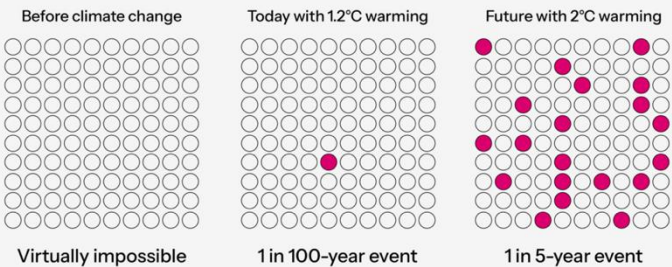


*Truncated axis

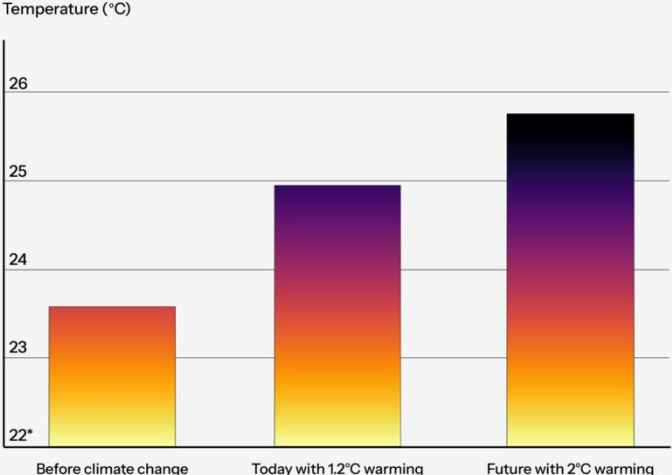




How often should we expect similar October heatwaves in Madagascar?



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*Truncated axis

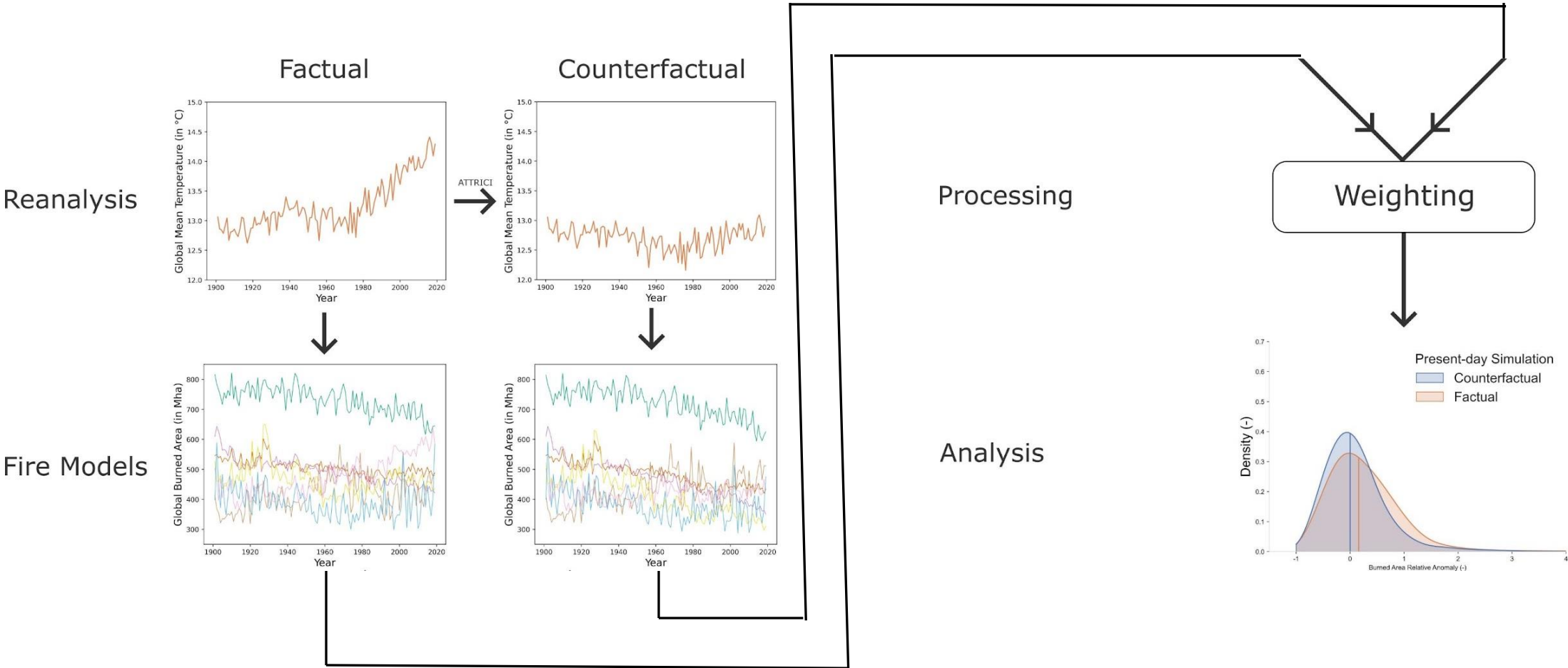


- What is the overarching result?
- Why is it important to use more than one line of evidence?
- How can you do that?
- What data sources are available for attribution studies?
- How do they differ for different regions of the world?
- How does the confidence in attribution (& projection) results differ depending on the type of hazard?



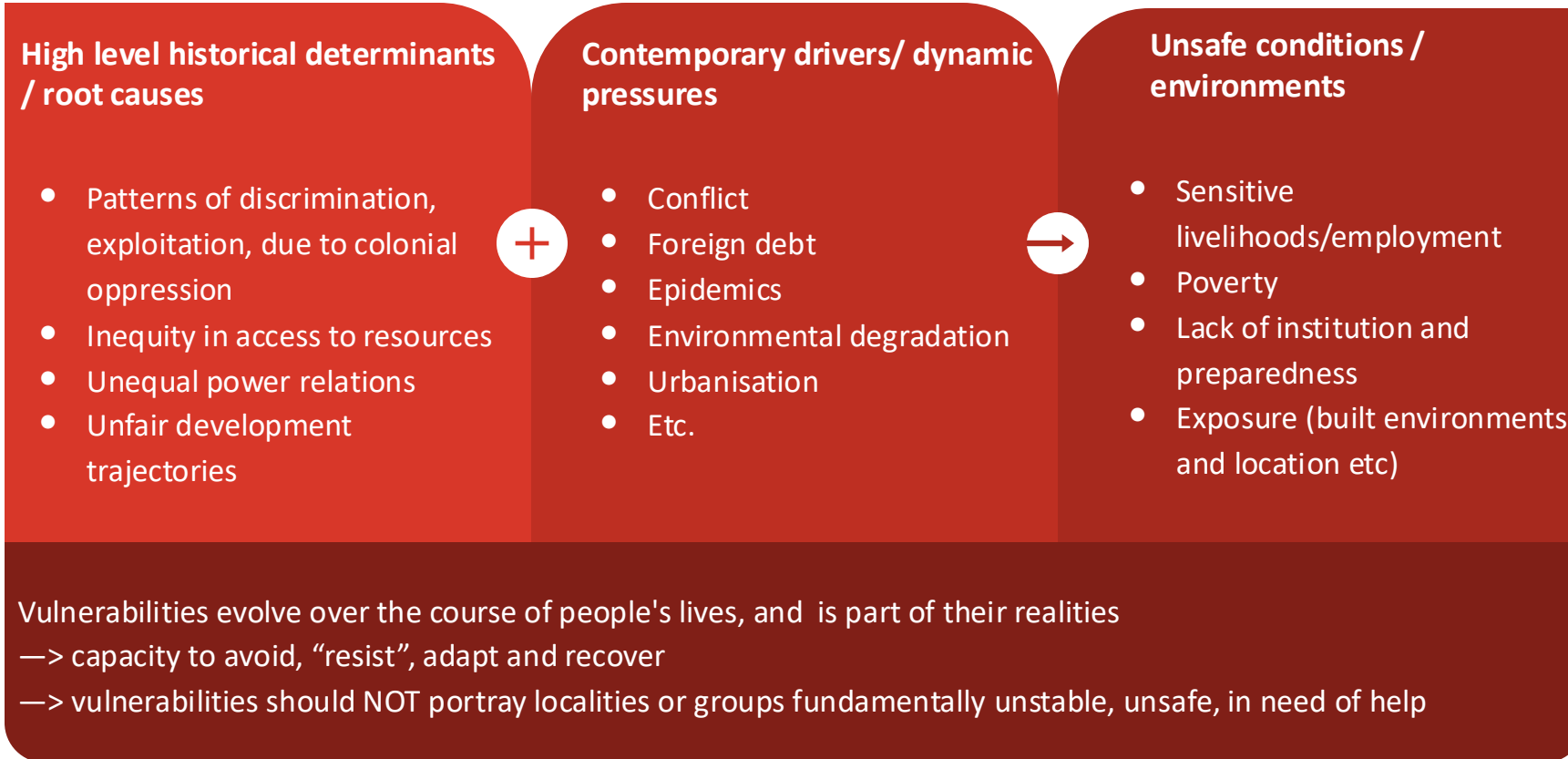
Impact attribution (lecture 2, 7 & 9)

why is our confidence much different?



Who is Vulnerable? Reasons for vulnerability?

(lectures 2 & 3)





Who is Vulnerable ?

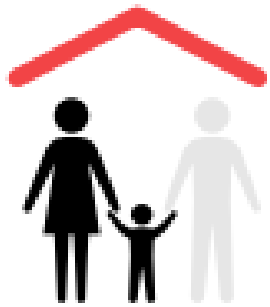
Categorisation in “women, elderly, sick” is too absolut, generic and homogenous

Creates limited ways in which people are categorised and understood

Misses the “uncategorised”

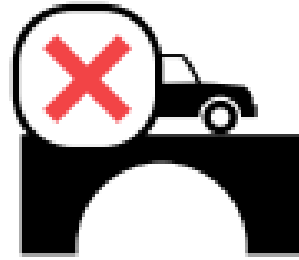
Conceals social & historical factors that shape the underlying vulnerabilities

→ Vulnerabilities intersect in complex ways and differ within communities, across societies.



Out of sight

The people we fail to see



Out of reach

The people we can't get to



Out of the loop

The people we unintentionally exclude



Out of money

The people we don't prioritize



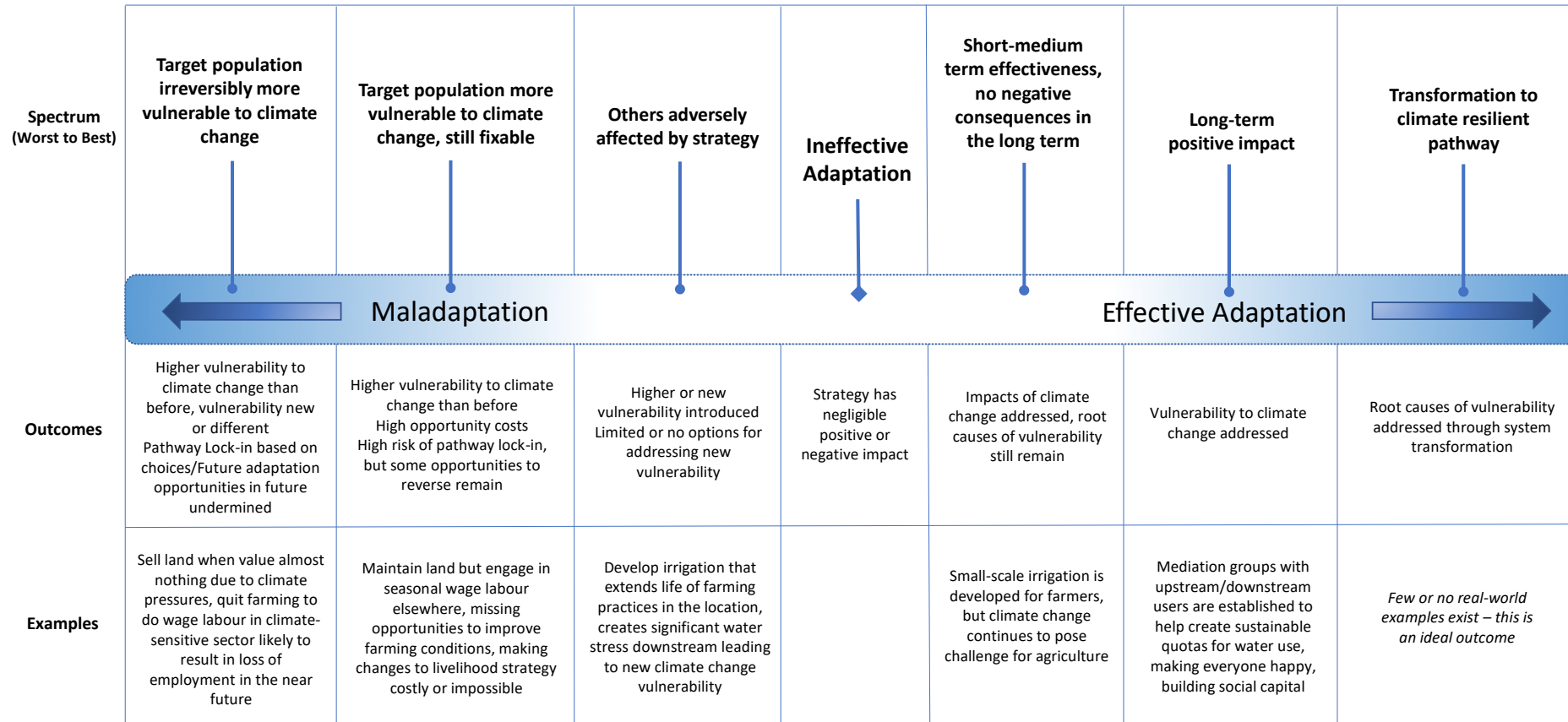
Out of scope

The people who 'aren't our problem'

- What are vulnerable groups?
- Why is a functional understanding of vulnerability more useful?
- Are they more useful in all circumstances?
- How can one quantify vulnerability?
- What do you miss when quantifying uncertainty on a national scale?
- Give examples of people out of sight, out of reach....
- How can people be brought in sight? Scope?
- What do you need to consider when communication climate change issues with specific groups?

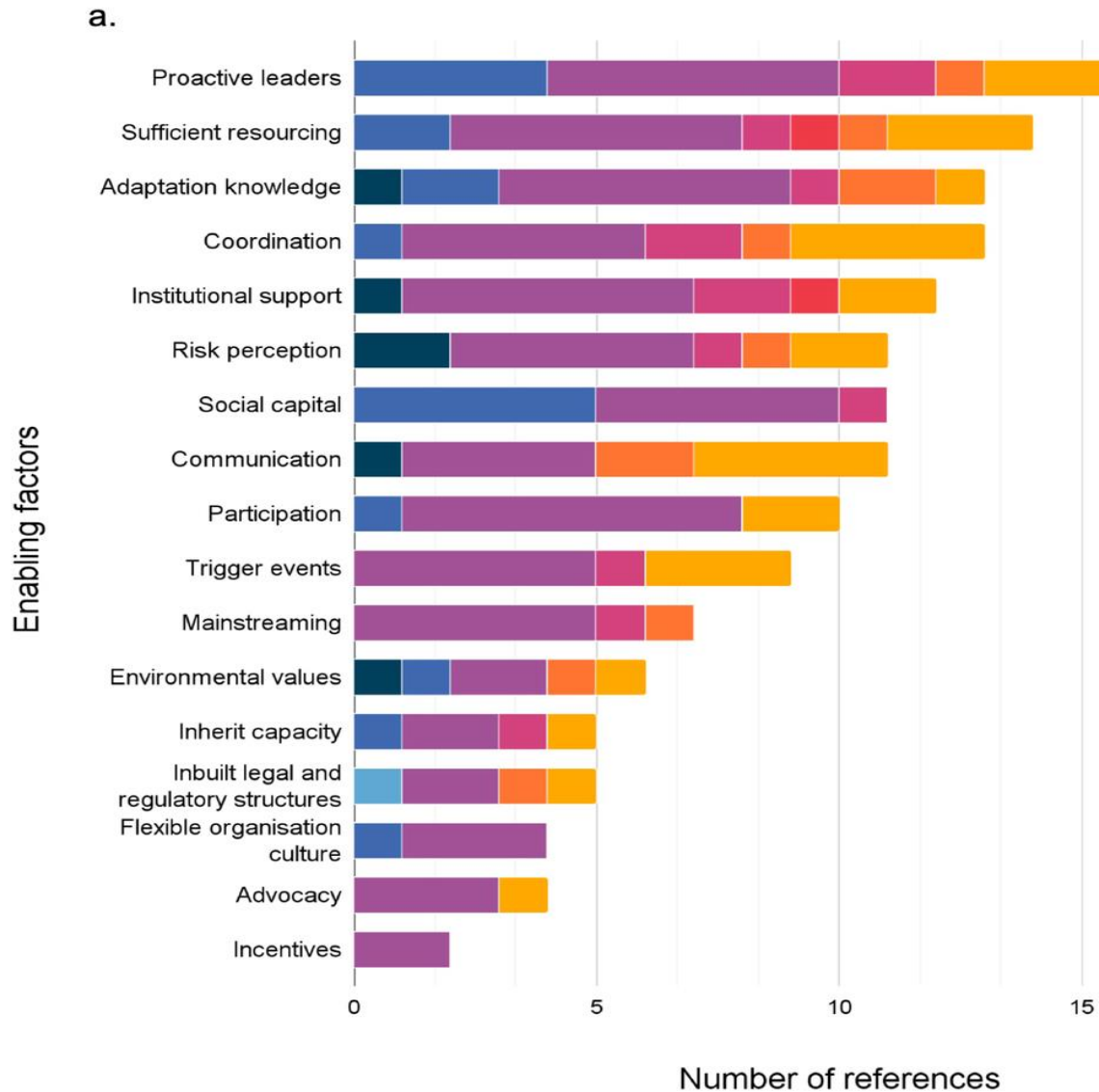


Maladaptation – Incremental - transformational

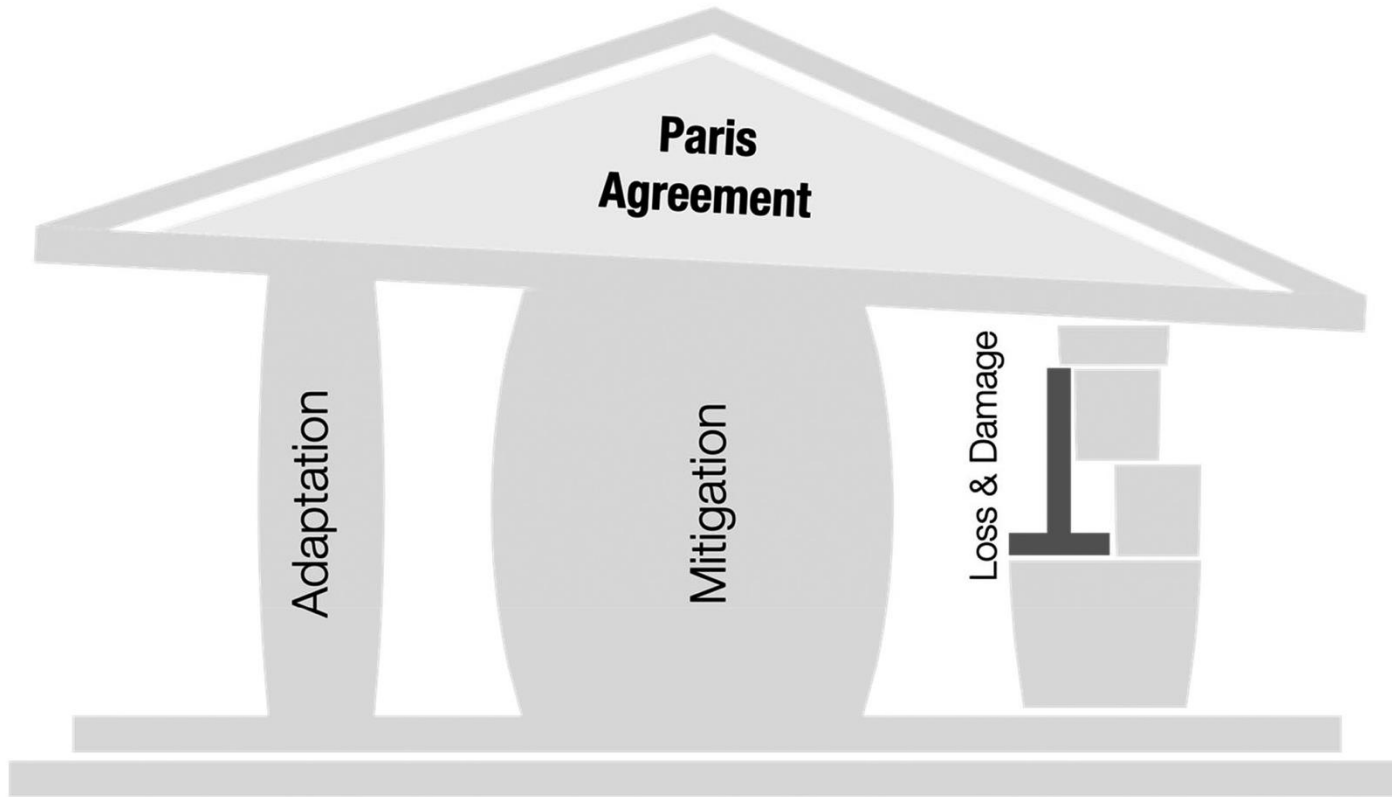




Lecture 6 (& 8)



- What is transformational adaptation?
- What is maladaptation?
- Give examples for different adaptation options for tropical cyclones, floods, heat
- What are barriers to adaptation?
- What are enablers for adaptation?
- What are examples for either?
- When you think of a coastal country, what are key hazards?
- A landlocked country? A global North city? A global South rural area?
- What are key hazards wrt human health? Food security?



- Why is L&D not defined?
- What are obstacles with respect to addressing L&D under UNFCCC?
- Why is L&D important from a justice perspective?



- Why is climate litigation on the rise?
- How have recent judgements/advisory opinions used scientific evidence?
- Why is scientific causation important for legal causation?
- Why is climate litigation not a panacea?
- What is the normative superstructure?



Questions? Office hours?